

The Exact Minimal Time for One-End Observation and Control of a Finite String

Route-A source-local certificate HCS-C287

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Abstract

We determine the sharp one-end observation and boundary-control time for the one-dimensional Dirichlet wave equation. At that exact time the endpoint normal derivative satisfies a Parseval identity with constant $4/c^3$, giving observability including the critical equality. Every shorter window misses a nonzero smooth periodic traveling profile, so the threshold is necessary as well as sufficient. Hilbert Uniqueness Method duality yields the matching one-end Dirichlet control theorem on the precise transposition space. Independent exact evidence audits constants without replacing the infinite-dimensional proof.

1 Frozen string and its full revival

Let $L, c > 0$ and consider

$$u_{tt} - c^2 u_{xx} = 0, \quad 0 < x < L, \quad u(0, t) = u(L, t) = 0. \quad (1)$$

The adjoint data lie in $H_0^1(0, L) \times L^2(0, L)$, with conserved energy

$$E(t) = \frac{1}{2} \int_0^L (|u_t(x, t)|^2 + c^2 |u_x(x, t)|^2) dx. \quad (2)$$

The observation is $u_x(L, \cdot)$. There is no zero mode: the frequencies are $\omega_n = n\pi c/L$, $n \geq 1$.

Lemma 1 (Least full-space revival). *The least positive time for which the wave group is the identity on the full energy space is*

$$T_* = \frac{2L}{c}.$$

Proof. At time T_* , every phase $e^{\pm i\omega_n T_*} = e^{\pm 2\pi i n}$ is one. Conversely, an identity time must return the $n = 1$ mode and is therefore an integer multiple of $2L/c$. In particular, L/c multiplies the n th mode by $(-1)^n$ and is not the full-space identity. $\square$

2 Critical endpoint Parseval identity

Every finite-energy Dirichlet solution has the periodic d'Alembert form

$$u(x, t) = F(x + ct) - F(-x + ct), \quad F(s + 2L) = F(s), \quad F' \in L^2(\mathbb{R}/2L\mathbb{Z}). \quad (3)$$

Constants in F are irrelevant. Differentiating gives

$$u_x(L, t) = 2F'(L + ct), \quad E = c^2 \int_0^{2L} |F'(s)|^2 ds. \quad (4)$$

The energy equality follows by adding the squares of u_t and cu_x and changing variables over the two reflected halves.

Theorem 1 (Exact critical observability). *Every solution of (1) satisfies*

$$\int_0^{2L/c} |u_x(L, t)|^2 dt = \frac{4}{c^3} E(0). \quad (5)$$

Consequently one-end boundary observability holds for every $T \geq 2L/c$, including equality.

Proof. At the critical time, $s = L + ct$ traverses one complete period. Hence

$$\int_0^{2L/c} |u_x(L, t)|^2 dt = \frac{4}{c} \int_0^{2L} |F'(s)|^2 ds = \frac{4}{c^3} E(0).$$

For a longer interval, integrate the nonnegative observation over its first critical subinterval. $\square$

For comparison, if

$$u(x, t) = \sum_{n \geq 1} \left(a_n \cos \omega_n t + \frac{b_n}{\omega_n} \sin \omega_n t \right) \sin \frac{n\pi x}{L},$$

orthogonality over $[0, 2L/c]$ gives (5) mode by mode. The periodic proof is stronger for the sharpness argument below because it does not truncate the solution.

3 Every shorter window misses a smooth wave

Theorem 2 (Sharp failure below the round trip). *If $0 \leq T < 2L/c$, a nonzero smooth solution of (1) exists with*

$$u_x(L, t) = 0 \quad (0 \leq t \leq T).$$

Thus no observability inequality, and hence no dual exact one-end control, holds before $2L/c$.

Proof. On the circle $\mathbb{R}/2L\mathbb{Z}$, the observed arc $\{L + ct : 0 \leq t \leq T\}$ has length $cT < 2L$. Its complement contains an open arc. Choose two smooth bumps of opposite integrals inside that complement; their sum is a nonzero smooth mean-zero function G . Let $F' = G$ and take a periodic primitive. Equations (3)–(4) give a nonzero smooth energy solution whose endpoint trace vanishes on the whole observation window. $\square$

This is an infinite-dimensional obstruction. A finite Galerkin Gramian can be nonsingular on a shorter interval and cannot replace Theorem 2.

4 HUM duality and function spaces

For the controlled state, keep $y(0, t) = 0$ and prescribe the Dirichlet value $y(L, t) = h(t)$. With $h \in L^2(0, T)$, solutions are understood by transposition in

$$L^2(0, L) \times H^{-1}(0, L).$$

Green duality pairs the boundary control with $c^2 u_x(L, t)$. Define the control-adjoint observation by $\mathcal{O}_T(u_0, u_1) = c^2 u_x(L, \cdot)$. At $T = 2L/c$, Theorem 1 is the norm identity $\|\mathcal{O}_T(u_0, u_1)\|_{L^2(0, T)}^2 = 4cE(0)$: thus $\mathcal{O}_T$ is a scaled isometry onto its closed range, not a claimed surjection onto all of $L^2(0, T)$. The HUM Gramian $\mathcal{O}_T^* \mathcal{O}_T$ is therefore coercive and, after the energy-space Riesz identification, an isomorphism onto the dual state space. Time reversibility gives exact control between arbitrary states in this dual space for every $T \geq 2L/c$; Theorem 2 proves failure below it.

HUM was introduced systematically by Lions [1]. Sharp boundary geometric control for waves is owned by Bardos, Lebeau, and Rauch [2]; in one dimension characteristics encode the ray condition. We claim the explicit convention-complete identity and executable boundary audit here, not priority for HUM or geometric control.

5 Executable receipt and Route-A boundary

The retained receipt has 16 positive (L, c) rows, 256 exact modal cells, 16 revival cells, and 16 strict subcritical arcs. A checker derived from the periodic coordinate reports 2,804 assertions; a separate symbolic route reports 86 identities; two isolated outputs are byte identical; and 23/23 repaired/stale-hash attacks fail. These cells audit (5); they do not prove infinite-dimensional observability or HUM.

The wave group has nonisolated periodic families, not an arithmetic primitive ledger. The Dirichlet Laplacian gives a natural source quantization but no rational-prime carrier, prime-power amplitude, target determinant, or same-clock target bridge. Under NO_BAD_EULER_OR_ROOT_NUMBER, the tuple is

$$(A0_FAIL, A1_WEAK, A2_FAIL, A3_FAIL, A4_NATURAL_QUANTIZATION),$$

the overall verdict is ROUTE_A_REJECTED, and Route B is disabled.

References

- [1] J.-L. Lions, *Exact controllability, stabilization and perturbations for distributed systems*, SIAM Review **30**(1) (1988), 1–68, [doi:10.1137/1030001](https://doi.org/10.1137/1030001).
- [2] C. Bardos, G. Lebeau, and J. Rauch, *Sharp sufficient conditions for the observation, control, and stabilization of waves from the boundary*, SIAM Journal on Control and Optimization **30**(5) (1992), 1024–1065, [doi:10.1137/0330055](https://doi.org/10.1137/0330055).